INCH-POUND
A-A-55014B
April 15, 2020
SUPERSEDING
A-A-55014A
July 27, 2001

COMMERCIAL ITEM DESCRIPTION

TORCH OUTFITS, WELDING, GAS TUNGSTEN ARC WELDING (GTAW) AND COMPONENT PARTS

The General Services Administration has authorized the use of this commercial item description, for all federal agencies.

1. SCOPE. This commercial item description (CID) covers complete welding torch outfits intended for use in gas tungsten arc welding (GTAW), also commonly referred to as tungsten inert gas (TIG) welding.

2. CLASSIFICATION. The gas tungsten arc welding torch outfits shall be of the following types and sizes. The type and size to be furnished shall be as specified (see 7.3(b)). Torch outfit types and sizes conform to the classification of torch types and sizes of A-A-51187. "Welding Torches and Component Parts, Gas Tungsten Arc Welding (GTAW)".

Type I - Air cooled
Type II - Water cooled

Size - Sizes are shown in tables I and II

TABLE I. Sizes of type I air-cooled GTAW torch outfits.

Size	Current Rating (A)
080	80
100	100
125	125
150	150
200	200

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TABLE II. Sizes of type II water-cooled GTAW torch outfits.

Size	Current Rating (A)
180	180
200	200
250	250
300	300
350	350
500	500

3. SALIENT CHARACTERISTICS.

3.1 General Requirements. The torch outfit shall be provided as a complete outfit that is ready for use when connected to a gas source, power source, and, if a type II torch, a water source. Unless otherwise specified (see 7.3(c)), torch outfits manufactured to metric dimensions shall be considered if they meet dimensional specifications, cable and hose connector standards, and other requirements stated herein.

3.2 Components. The torch outfit shall include, as a minimum, the following components:

3.2.1 Torch. The type I torch, power cable, and gas hose and the type II torch, power cable, gas hose, and water hoses shall conform to the requirements of A-A-51187.

3.2.2 Power cable adapter. Unless otherwise specified (see 7.3(d)), a power cable adapter compatible with the torch shall be provided.

3.2.3 Ground cable and clamp assembly. If required, a ground cable (15 or 25 feet) and clamp assembly shall be provided as specified (see 7.3(e)).

3.2.4 Pressure regulating system. Unless otherwise specified (see 7.3(f)), the torch outfit shall have a gas pressure regulating system that is assembled as a single unit and includes, as a minimum, a pressure regulator, inlet pressure gage, pressure relief valve, and flowmeter. The inlet connector to the system shall conform to Compressed Gas Association (CGA) V-1, "Standard for Compressed Gas Cylinder Valve Outlet and Inlet Connections", connection number 580. Unless otherwise specified (see 7.3(g)), the system shall be suitable for use with argon and helium gases.

3.2.4.1 Pressure regulator. Unless otherwise specified (see 7.3(h)), the pressure regulator shall be either one or two stage. The regulator shall conform to the requirements of CGA E-4, "Standard for Gas Pressure Regulators". The inlet pressure rating of the regulator shall not be less than 3,000 pounds per square inch gage. The regulator shall have a 60 micron, or finer filter, made of porous bronze or other corrosion-resistant metal and located in the inlet ahead of the gas seat. The discharge pressure of the regulator shall be preset to a level that is compatible with the pressure rating of the flowmeter.

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3.2.4.2 Inlet pressure gage. The pressure regulating system shall be equipped with an inlet gas pressure gage. The gage shall conform to the requirements of American Society of Mechanical Engineers (ASME) B40.100, "Pressure Gauges and Gauge Attachments", meeting the accuracy requirements for grade B. Unless otherwise specified (see 7.3(i)), the pressure gage range shall be 0 to 4,000 pounds per square inch gage.

3.2.4.3 Pressure relief valve. A self-resetting pressure relief valve compatible with the pressure rating of the flowmeter shall be provided at the outlet of the pressure regulator.

3.2.4.4 Flowmeter. The flowmeter shall have a safe operating pressure rating of not less than 75 pounds per square inch gage. Unless otherwise specified (see 7.3(j)), the flowmeter shall be a direct reading type and be calibrated in cubic feet per hour for argon and helium gases. If the flowmeter provided is calibrated for both gases, the correct scale for each gas shall be clearly identified. Unless otherwise specified (see 7.3(k)), the operating flow range shall not be less than 10 through 40 cubic feet per hour for argon and 30 through 150 cubic feet per hour for helium within an accuracy of ±10 percent of full scale. The outlet connector shall conform to the requirements of CGA E-1, "Standard for Rubber Welding Hose and Hose Connections for Gas Welding, Cutting, and Allied Processes", connection number 032.

3.2.4.5 Gas supply hose. When specified (see 7.3(l)), a gas supply hose shall be provided to connect the flowmeter and the power source. Unless otherwise specified (see 7.3(m)), the length of the hose shall be 25 feet. The hose shall have a minimum pressure rating of 150 pounds per square inch gage and shall conform to the requirements of the Association for Rubber Products Manufacturers (ARPM) IP-7, "Specifications for Rubber Welding Hose", for rubber welding hoses. The ends of the hose shall have hose fittings which conform to the requirements of CGA E-l, connection number 032.

3.2.5 Cooling water circulator. When specified (see 7.3(n)), a portable, self-contained, closed-loop cooling water circulator shall be provided, which has the cooling capacity to maintain the specified type II torch at the manufacturer's recommended safe operating temperature. The water circulator shall be complete and ready to use. Unless otherwise specified (see 7.3(o)), the water circulator shall be electrically powered and shall operate from a 115-volt, single-phase, 60-hertz circuit.

3.2.5.1 Water hoses. When specified (see 7.3(p)), the type II torch outfit shall have a water supply and a water drain hose with fittings for connection to the power cable adapter. Unless otherwise specified (see 7.3(q)), the length of the hoses shall be 12.5 feet.

3.2.5.2 Overheating prevention device. If required, the type II torch outfits shall be provided with an overheating prevention device as specified (see 7.3(r)). The device shall shut off current to the torch if overheating occurs from a reduction in cooling water flow or from torch operation at higher than its rated capacity.

3.2.6 Optional accessories. Optional accessories shall be furnished as specified (see 7.3(s)).

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3.2.7 Replacement parts or components. If required, replacement parts shall be furnished as specified (see 7.3(t)). Each outfit replacement part or component shall be a replacement for the brand name and part number specified (see 7.3(u)). The components furnished are not required to be manufactured by a specific manufacturer, but the components provided shall mate with and perform without modification of either the replacement part or the component it is mating with.

3.3 System of units. The U.S. Customary System of Units (US) or the International System of Units (SI) shall be used to graduate measuring and indicating devices. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 7.3(v)). Regardless of the measurement system used, all measuring and indicating devices on the torch outfit shall be graduated in the same system. When specified (see 7.3(w)), measuring and indicating devices shall be graduated in both the US and SI system of measurements.

4. REGULATORY REQUIREMENTS. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

5. PRODUCT CONFORMANCE.

5.1 Product Conformance. The products provided shall meet the salient characteristics of this Commercial Item Description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance.

5.2 Market Acceptability. The product offered must have been previously sold either to the government or on the commercial market.

5.3 Inspection requirements. The torch shall be inspected to determine compliance with all requirements specified in this CID.

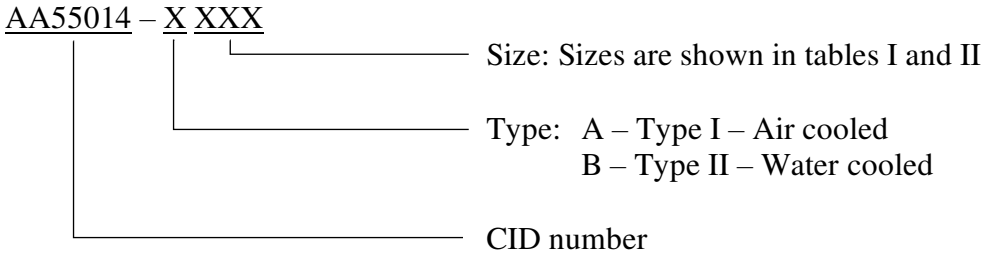
5.4 Warranty. Unless otherwise specified (see 7.3(x)), the manufacturer's standard commercial warranty, terms, and conditions shall apply.

6. PACKAGING. Unless otherwise specified (see 7.3(y)), preservation, packing, and marking shall be as specified in the contract or order.

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7. NOTES.

7.1 Part or identification number (PIN). The following part or identification numbering procedure is for government purposes and does not constitute a requirement for the contractor.



7.2 Sources of documents.

7.2.1 Government documents. Copies of the FAR may be obtained from https://www.acquisition.gov/far/. Electronic copies of commercial item descriptions may be obtained from https://quicksearch.dla.mil/.

7.2.2 ASME standards. Copies of ASME standards may be obtained from the American Society of Mechanical Engineers, https://www.asme.org/.

7.2.3 CGA standards. Copies of CGA standards may be obtained from the Compressed Gas Association, https://www.cganet.com/.

7.2.4 ARPM standards. Copies of ARPM standards may be obtained from the Association for Rubber Products Manufacturers, https://arpminc.com/.

7.3 Ordering data. Purchaser should specify the following:

(a) CID title, number, date, and CID PIN.
(b) Type and size required (see 2).
(c) Metric dimensions acceptable, if different (see 3.1).
(d) Power cable adapter, if different (see 3.2.2).
(e) Ground cable and clamp assembly, as specified (see 3.2.3).
(f) Pressure regulating system, if different (see 3.2.4).
(g) Gas types, if different (see 3.2.4).
(h) Pressure regulator, if different (see 3.2.4.1).
(i) Pressure gage range, if different (see 3.2.4.2).
(j) Flowmeter reading and gas calibration, if different (see 3.2.4.4).
(k) Flowmeter operating flow range, if different (see 3.2.4.4).
(l) Gas supply hose, if required (see 3.2.4.5).
(m) Gas supply hose length, if different (see 3.2.4.5).
(n) Type II torch cooling water circulator, if required (see 3.2.5).
(o) Type II torch cooling water circulator electrical power, if different (see 3.2.5).
(p) Type II torch water hoses, if required (see 3.2.5.1).

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(q) Type II torch water hose length, if different (see 3.2.5.1).
(r) Type II torch overheating prevention device, as specified (see 3.2.5.2).
(s) Optional accessories, as specified (see 3.2.6).
(t) Replacement parts, as specified (see 3.2.7).
(u) Brand name and part number of replacement part, as specified (see 3.2.7).
(v) System of units required (see 3.3).
(w) Dual system of units, if required (see 3.3).
(x) Warranty, if different (see 5.4).
(y) Packaging requirements, if different (see 6).

MILITARY INTERESTS:

Custodians:
Army-AR
Navy-SH
DLA-GS

Reviewer:
Army-CE, CR4
Navy-MC, YD

CIVIL AGENCY
COORDINATING ACTIVITY:
GSA-FAS

Preparing Activity:
DLA-GS6

(Project 3431-2020-002)

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